

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Tune and evaluate LLMs for production agentic systems using prompt engineering and retrieval-augmented generation, combining GraphRAG with vector-based RAG for grounded retrieval.
- Own multi-step agent workflows end-to-end (LangGraph, CrewAI, AutoGen, OpenAI Agents SDK) with tool-use integrations for natural-language querying over structured enterprise data.
- Built a deployment-gated evaluation framework covering model quality, hallucination, and regression across LLM/agent outputs, enforcing guardrails before production release.
- Write, test, and maintain production-quality Python and TypeScript services, working in Agile sprints with Git-based version control.

QureAI (AI Startup)

Mar 2026 – Present

ML Infrastructure Engineer

- Developed deep learning models (PyTorch) with attention mechanisms and domain-specific architectures; ran hands-on model training/fine-tuning on multi-GPU clusters with compression and quantization.
- Built custom evaluators for RAG/agent workflows tracking drift, latency, and regression across model versions, catching failures before they reached production.
- Improved semantic retrieval recall 61% to 89% on a 2K-query eval set by rebuilding the chunking pipeline with hybrid dense-sparse retrieval and re-ranking.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production C++ and SQL-backed microservices for 1M+ users in the financial services domain, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly financial data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Standardized failure-handling contracts (retry, timeout, propagation semantics) across 25+ services, reducing production escalations 60% in one quarter.

Projects

LangFetch | tinyurl.com/LangFetch

- LangGraph agent with MCP-compatible tool-use performing semantic parsing of natural-language questions into dependency-ordered SQL subqueries with Pinecone vector retrieval and hybrid re-ranking.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production LLM evaluation platform with hallucination detection, drift tracking, rubric-based scoring, and regression alerting across model versions; FastAPI backend, PostgreSQL.

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, TypeScript, SQL, Java, C++

LLM & NLP: PyTorch, LLM Fine-Tuning (Multi-GPU), Prompt Engineering, RAG, Semantic Parsing, Model Compression/Quantization

Evaluation & Safety: LLM Evaluation, Hallucination Detection, Regression Tracking, Guardrails, Responsible AI

Cloud & Platform: AWS, GCP, Docker, Kubernetes, PostgreSQL, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer